

Groups of order 12

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Let G be a group of order $12 = 2^2 * 3$. Let F be a subgroup of order 4 (a sylow 2) and T a sylow 3.

Problem 1. Use the sylow theorems, and the "rose petal" picture to argue that either F or T is normal.

Proof. By the 3rd Sylow Theorem, the number of sylow 2 subgroups $n_2|3$ and $n_2 \mod 2 = 1$ so $n_2 = 1$ or $n_2 = 3$. If $n_2 = 1$, so F is normal. If $n_2 = 3$, then by the 3rd Sylow Theorem, the number of sylow 3 subgroups $n_3|4$ and $n_3 \mod 3 = 1$, $n_3 = 1, 4$. If $n_3 = 1$, T is normal. If $n_3 = 4$, then we have 4 sylow 3 subgroups which are cyclic (3 is prime) and thus only share 1 element the identity. So the number of order 3 elements in G is $4(2) = 8$. Every thing in a sylow 2 subgroup is not order 3. The number of non order 3 elements in G is 4. So there is only 1 sylow 2 subgroup and thus it is normal. $\square$

Problem 2. Show that H and K are subgroups of a group G , both normal, and $H \cap K = \{e\}$, then A and B commute (there is a one line proof). Conclude our order 12 G is Abelian if both F and T are normal

Proof. $hkh^{-1} \in K$ and $kh^{-1}k^{-1} \in H$ $hkh^{-1}k^{-1} \in H \cap K$ by closure. Thus $hkh^{-1}k^{-1} = e$ thus $hk = kh$.

Thus since $F \cap T$ is trivial if they both are normal they would commute. Then the order of FT is 12 thus because $F \times T \rightarrow G$ is a homomorphism. Thus $F \times T \cong G$. Because F has order 4 its elements have orders 1, 2, or 4. If there is a element with order 4, it is cyclic and thus is abelian. If there is a element with order 2, the group is $\mu_2 \times \mu_2$ which is abelian. If it has element with order 1, the element is the identity and thus must have another element with order 2 or 4. So all subgroups with order 4 are abelian. T is a abelian because it is a μ_3 . Thus G is abelian. $\square$

Problem 2.5. We first assume G is not Abelian, we will do the Abelian cases afterwards.

Say T is normal. Explain why $G = T \rtimes F$, where the ugly $\rtimes$ means semi-direct product (with the subgroup on the left being normal). All that remains is to consider the possible $C : F \rightarrow \text{Aut}(T)$. Explain why $\text{Aut}(T) = 1, -1$, where -1 is the automorphism $t \mapsto t^{-1}$.

Proof. $\text{Aut}(\mu_3) = (F_3)^*$ but $(F_3)^* = \{1, -1\}$. So $\text{Aut}(\mu_3) = \mu_2$ $\square$

Problem 3. Suppose F is cyclic of order 4. Say $F = \langle f \rangle$. Explain why $G = \langle f, t | f^4 = t^3 = 1, ft f^{-1} = t^{-1} \rangle$.

Proof. $F \times T \rightarrow G$ is injective because $F \cap T = e$. The map is surjective since $|FT| = |F||T|/|F \cap T| = 12/1 = 12$. So $FT = G$.

Because $T \rtimes_{t \mapsto t^{-1}} F = \langle t | t^3 = e \rangle \rtimes_{t \mapsto t^{-1}} \langle f | f^3 = e \rangle = \langle t, f | t^3 = t^4 = e, ft f^{-1} = t^{-1} \rangle$ $\square$

Problem 4. Now say $F = \mu_2 x \mu_2$. Consider the possible $C : F \rightarrow \text{Aut}(T)$. Explain why this is non-trivial (under our assumption that G is not Abelian). Now show you can choose a "basis" for F , a, b so that $C(a) = -1, C(b) = 1$. Then $G = \langle a, b, t \mid a^2 = b^2 = 1, ab = ba, ata = t^{-1}, btb = t \rangle$. Note F is iso to F_2^2 , "basis" means a basis of this vector space.

Proof. If C was trivial then F and T would commute thus making G be abelian. Let $a = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. $C(a) = -1$ and $C(b) = 1$, $C(a^2b) = 1$. Then $G = \langle a, b, t : a^2 = b^2 = t^3 = 1, ab = ba, ata = t^{-1}, btb = t \rangle$ $\square$

Problem 5. Now suppose F is normal. There are 2 possibilities for F . Say first F is cyclic. Explain why $\text{Aut}(F) = \{1, -1\}$, same meaning as before. Show that G is Abelian (so we ignore this case for now). Now say F is $\mu_2 x \mu_2$. Explain why $\text{Aut}(F) = \text{GL}_2(F_2) = S_4$.

Explain why $C : T \rightarrow \text{Aut}(F)$ must take a generator to a 3-cycle. Explain why by changing the generator we can make this either 3-cycle (so either matrix of order 3). Now show $G = \langle t, a, b \mid t^3 = a^2 = b^2 = 1, ab = ba, tat^{-1} = b, btb^{-1} = a \rangle$

Proof. If F is cyclic then it is μ_4 . $\text{Aut}(\mu_4) = (F_4)^* = \{1, 3\} = \{1, -1\}$. $F \cap T = \{e\}$. There is no map except the trivial map because elements in μ_3 have order 3 and elements in μ_2 have order 2.

Say $F \cong \mu_2 \times \mu_2 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = F_2^2$. Then F are 2 dimensional, vectors in F_2 . The automorphism of F_2 vectorspace are the invertible matrices. So $\text{Aut}(F) \cong \text{GL}_2(F_2) \cong S_3$.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$Y = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C_1(y) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} y \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^{-1} \text{ or } C_2(y) = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} y \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{-1}.$$

$$a = e_1, b = e_2, c = e_1 + e_2.$$

$$C_1(a) = b, C_1(b) = ab$$

$$F \times_{C_1} T = \langle a, b, X : a^2 = b^2 = e, ab = ba, XaX^{-1} = b, XbX^{-1} = ab \rangle$$

$\square$

Problem 7. So we have shown that a non-Abelian group of order 3 is one of three groups. Prove no two of these groups are isomorphic. We already know two Non-Abelian groups of order 12, A_4 and D_6 . Which is which?

Proof. None of the groups are isomorphic because they all have different sylow 4 subgroups and groups with the different sylow 4 subgroups must not be isomorphic.

$A_4 \cong \langle a, b, X : a^2 = b^2 = X^3 = e, ab = ba, XaX^{-1} = b, XbX^{-1} = ab \rangle$ where a and b are disjoint two cycles like (12)(34) and (14)(23) and X is the 3 cycle.

$D_6 \cong \langle a, b, t : a^2 = b^2 = t^3 = 1, ab = ba, ata^{-1} = t^{-1}, btb^{-1} = t \rangle$ where b is rotation with order 2, a is a reflection, and t is a rotation with order 3. $\square$

Problem 7. Now we consider Abelian groups of order 12. Next term we (happy few) will show that any finite Abelian group is iso to a product of cyclic groups. But here we can do the classification by hand. Show $G = F \rtimes T$ (ordinary product). and that there are two possibilities.

Proof. If G is abelian, then $FT = TF$ so $F \times T \rightarrow G$ is a homomorphism. Furthermore $|FT| = 12$ and $F \cap T = \{e\}$ so the homomorphism is bijection. F is either $\mu_2 \times \mu_2$ or F is μ_4 . So $G = \mu_2 \times \mu_2 \times \mu_3$ or $G = \mu_4 \times \mu_3 \cong \mu_{12}$ $\square$